Bearings

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Bearings Basics

1. 000°

2. 090°

3. 008°

4. C) 240

5. East

6. 180°

7. C) South

8. N, NE, E, SE, S, SW, W, NW

9. 045°

10. D) North-West

11. A, C and D are valid (B is not written with three digits; E exceeds a full turn of 360°)

12. 270°

13. They are exact opposites — 090° and 270° differ by 180°, so each is the back bearing of the other

14. B and C (150° and 200° lie between 090° and 270°; 060°, 300° and 350° do not)

15. A) North

Reading & Finding Bearings

16. North-East (a bearing of 050° lies between North, 000°, and East, 090°, closer to North-East, 045°)

17. 245° (065° + 180° = 245°)

18. 322° (142° + 180° = 322°)

19. C) 120° (300° − 180° = 120°)

20. 030° (210° − 180° = 030°)

21. A, C and D are true (B is false — 095° + 180° = 275°, not 185°; E is false — 300° − 180° = 120°, not 110°)

22. 75° (turning clockwise from North through 75° gives a bearing of 075°)

23. B) South-East (128° lies between East, 090°, and South, 180°, closer to South-East, 135°)

24. 010° (190° − 180° = 010°)

25. The back bearing points roughly South-West; adding 180° to 048° gives a back bearing of 228°

26. B) 178° (358° is 180° or more, so subtract 180°: 358° − 180° = 178°)

27. A, C and E are true (B is false — the back bearing is found by ADDING 180° when the bearing is less than 180°, not subtracting it; D is false — 265° is greater than 180°, not less)

28. 032° (212° − 180° = 032°)

29. 315° (North-West is a bearing of 315°)

30. 130° (310° − 180° = 130°)

Bearings Challenge

31. 240° (060° + 180° = 240°)

32. 252° (072° + 180° = 252°)

33. 308° (128° + 180° = 308°)

34. C) 120° (165° − 045° = 120°)

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35. 190° (100° + 90° = 190°)

36. A, B, D and E are true (C is false — since 155° is less than 180°, the back bearing is found by ADDING 180°, not subtracting it)

37. Corrected bearing = 220° (210° + 10°); back bearing = 040° (220° − 180°)

38. B) 70° (the angle between bearings 020° and 310°, measured the short way round, is 020° + (360° − 310°) = 020° + 050° = 070°)

39. A) C is the same point as A (265° is exactly the back bearing of 085°, since 085° + 180° = 265°, so the B-to-C leg exactly retraces the A-to-B leg)

40. 120° (300° − 180° = 120°)

41. A, B, D and E are true (C is false — 180° − 064° = 116°, which is not the back bearing; the correct back bearing is 064° + 180° = 244°)

42. 030° (210° − 180° = 030°)

43. D) North-West (128° + 180° = 308°, which is closest to 315°, i.e. North-West)

44. 6 km (260° is exactly the back bearing of 080°, since 080° + 180° = 260°, so C lies back along the same line towards the start; 10 km − 4 km = 6 km)

45. 180° (the back bearing is always the outward bearing plus 180°, so 250° − 070° = 180°)

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Homework Worksheet — Answers

Bearings Basics

1. 180°

2. 270°

3. 005°

4. C) 075

5. North

6. 000°

7. D) West

8. N, NW, W, SW, S, SE, E, NE

9. 225°

10. A) North-East

11. A, C and E are valid (B is not written with three digits; D exceeds a full turn of 360°)

12. 090°

13. They are exact opposites — 000° and 180° differ by 180°, so each is the back bearing of the other

14. A, D and E (030°, 320° and 355° are less than 090° or greater than 270°; 100° and 200° are not)

15. C) South

Reading & Finding Bearings

16. South-West (a bearing of 220° lies between South, 180°, and West, 270°, closer to South-West, 225°)

17. 215° (035° + 180° = 215°)

18. 018° (198° − 180° = 018°)

19. C) 260° (080° + 180° = 260°)

20. 335° (155° + 180° = 335°)

21. A, C and D are true (B is false — 100° + 180° = 280°, not 190°; E is false — 060° + 180° = 240°, not 260°)

22. 112° (turning clockwise from North through 112° gives a bearing of 112°)

23. C) North-West (302° lies between North, 360°/000°, and West, 270°, closer to North-West, 315°)

24. 160° (340° − 180° = 160°)

25. The back bearing points roughly North; adding 180° to 162° gives a back bearing of 342°

26. C) 185° (005° + 180° = 185°)

27. A, C, D and E are true (B is false — since 205° is 180° or more, the back bearing is found by SUBTRACTING 180°, not adding it)

28. 145° (325° − 180° = 145°)

29. 135° (South-East is a bearing of 135°)

30. 268° (088° + 180° = 268°)

Standard / Advanced

31. 280° (100° + 180° = 280°); this is closest to West (270°)

32. 150° (190° − 040° = 150°), which is more than a right angle (90°)

33. 205° (115° + 90° = 205°); this is closest to South-West (225°)

34. Corrected bearing = 165° (140° + 25°); back bearing = 345° (165° + 180°, since 165° is less than 180°)

35. 252° (072° + 180° = 252°), which is less than 270°

36. A, C, D and E are true (B is false — a bearing of 210° lies roughly to the south-west, so the pond is to the south-west of the barn, not the north-east)

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37. C) 90° (040° + (360° − 310°) = 040° + 050° = 090°)

38. A) They are the same point (100° + 180° = 280°, which exactly matches the B-to-C bearing)

39. 255° (075° + 180° = 255°)

40. A, B, D and E are true (C is false — 180° − 152° = 028°, which is not the back bearing; the correct back bearing is 152° + 180° = 332°)

41. E) North-West (135° + 180° = 315°, which is exactly North-West)

42. 342° (162° + 180° = 342°); this is closest to North (000°)

43. 7 km (280° is exactly the back bearing of 100°, since 100° + 180° = 280°, so C lies back along the same line towards the start; 12 km − 5 km = 7 km)

44. 065° + 180° = 245°, which matches the student's claim — so yes, the student is correct

45. Back bearing = 275° (095° + 180° = 275°); the two bearings differ by 180°

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